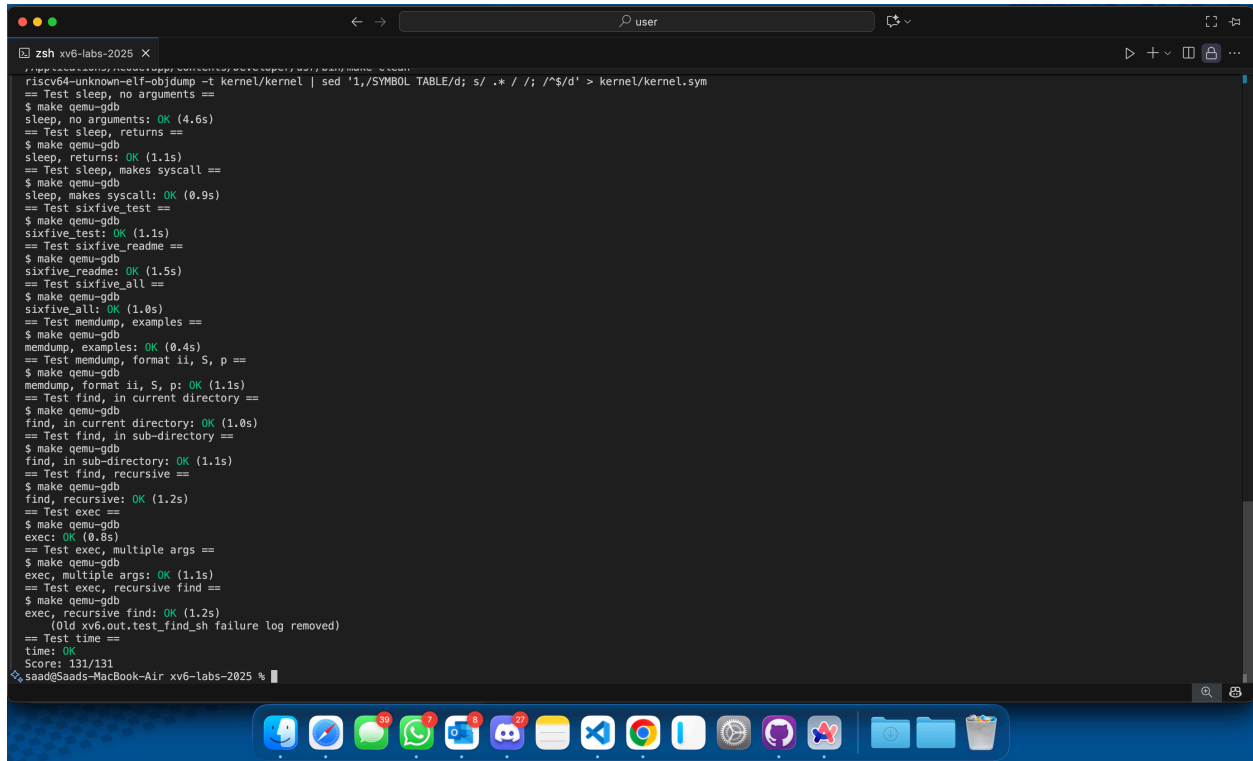


Xv6 Labs – Saad Inam 29068

problemSet1 Results (make grade)



```
zsh xv6-labs-2025 %  
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym  
== Test sleep, no arguments ==  
$ make qemu-gdb  
sleep, no arguments: OK (4.6s)  
== Test sleep, returns ==  
$ make qemu-gdb  
sleep, returns: OK (1.1s)  
== Test sleep, makes syscall ==  
$ make qemu-gdb  
sleep, makes syscall: OK (0.9s)  
== Test sixfive_test ==  
$ make qemu-gdb  
sixfive_test: OK (1.1s)  
== Test sixfive_readme ==  
$ make qemu-gdb  
sixfive_readme: OK (1.5s)  
== Test sixfive_all ==  
$ make qemu-gdb  
sixfive_all: OK (1.0s)  
== Test memdump, examples ==  
$ make qemu-gdb  
memdump, examples: OK (0.4s)  
== Test memdump, format ii, S, p ==  
$ make qemu-gdb  
memdump, format ii, S, p: OK (1.1s)  
== Test find, in current directory ==  
$ make qemu-gdb  
find, in current directory: OK (1.0s)  
== Test find, in sub-directory ==  
$ make qemu-gdb  
find, in sub-directory: OK (1.1s)  
== Test find, recursive ==  
$ make qemu-gdb  
find, recursive: OK (1.2s)  
== Test exec ==  
$ make qemu-gdb  
exec: OK (0.8s)  
== Test exec, multiple args ==  
$ make qemu-gdb  
exec, multiple args: OK (1.1s)  
== Test exec, recursive find ==  
$ make qemu-gdb  
exec, recursive find: OK (1.2s)  
    (Old xv6.out.test_find_sh failure log removed)  
== Test time ==  
time: OK  
Score: 131/131  
saad@Saads-MacBook-Air xv6-labs-2025 %
```

```
saad@Saads-MacBook-Air xv6-labs-2025 % make qemu
ballocc: first 1110 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.
lk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep 10
$ sixfive sixfive.txt
5
100
18
6
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
BA0
1819438967
100
z
xyzy
Example 5:
hello
w
o
r
l
d
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
64616564
$ echo > b
```

```
❖ saad@Saads-MacBook-Air xv6-labs-2025 % make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt use
ser/_usertests user/_grind user/_wc user/_zombie user/_logstre
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap
ballocc: first 1110 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/ke
lk-device,drive=x0,bus=virtio-mmio-bus.0
```

xv6 kernel is booting

```
hart 2 starting
hart 1 starting
init: starting sh
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ find . wc -exec echo hi
hi ./wc
$ sh < findtest.sh
$ mkdir: a failed to create
$ $ $ $ $ hello
hello
hello
$ $ █
```

Optional Exercises

```
❖ saad@Saads-MacBook-Air xv6-labs-2025 % make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_e
ser/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forph
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) bloc
ballocc: first 1151 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -
lk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ find . ^s.*
./sixfive.txt
./sh
./stressfs
./sleep
./sixfive
$ uptime
System uptime: 75 ticks
$ history
find . ^s.*
uptime
$ echo echo world >> testsh.txt
$ echo echo hello >> testsh.txt
$ sh < testsh.txt
hello
$ sleep 10
```